

Cylindrical Pullback Positive Cases for the Aronszajn-Null Game

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Source and Open Question

The source paper is Jakub Duda and Boaz Tsaban, *Null sets and games in Banach spaces*, arXiv:math/0601556, later published in *Topology and its Applications* **156** (2008), 56–60. For a separable real Banach space X , and a non-zero direction $x \in X$, let $\mathcal{A}_X(x)$ denote the Borel sets $E \subset X$ whose intersection with every affine line parallel to x has one-dimensional Lebesgue measure zero.

Duda–Tsaban define the Aronszajn-null game G_A for a Borel set $A \subset X$: Player I plays directions x_n , subject to the requirement that the sequence is dense in X , and Player II answers with $A_n \in \mathcal{A}_X(x_n)$. Player II wins if $A \subset \bigcup_n A_n$. They prove that if Player I has no winning strategy, then A is Aronszajn-null, and ask whether the converse holds:

$$A \text{ Aronszajn-null} \implies \text{Player I has no winning strategy in } G_A.$$

Result

Theorem 1 (Pullback of II-winning sets). *Let $T : X \rightarrow Y$ be a continuous linear surjection between separable real Banach spaces. Let $B \subset Y$ be Borel. If Player II has a winning strategy in the Aronszajn-null game on B in Y , then Player II has a winning strategy in the Aronszajn-null game on $T^{-1}(B)$ in X .*

Corollary 1 (Finite-dimensional cylindrical null sets). *Let X be a separable real Banach space, let $T : X \rightarrow F$ be a continuous linear surjection onto a finite-dimensional real Banach space, and let $N \subset F$ be Borel Lebesgue null. Then Player II has a winning strategy in the Aronszajn-null game on $T^{-1}(N)$ in X .*

Consequently, every countable union of such finite-dimensional cylindrical null sets is also II-winning.

Proof Intuition

The finite-dimensional packet proved that Lebesgue-null Borel sets in finite dimensions are II-winning. The point of this packet is that II-winning is stable under linear quotients. During a play in X , Player II projects Player I's directions to Y , ignores the zero projections, and lets the winning strategy in Y choose the projected response. Pulling the response back through T is legal in X : along a line parallel to x , the map T is a nonconstant affine map onto a line parallel to Tx , so null sections remain null after pullback.

Proof

Let Φ be a winning strategy for Player II in the game on $B \subset Y$. We define a strategy Ψ for Player II in the game on $T^{-1}(B) \subset X$.

Suppose Player I has played $x_n \in X$ at inning n . If $Tx_n = 0$, let Ψ play $\emptyset$. This is legal for every non-zero direction in X . If $Tx_n \neq 0$, feed Tx_n to the strategy Φ , together with the previous nonzero projected directions and projected responses. If Φ returns $C_n \in \mathcal{A}_Y(Tx_n)$, let

$$\Psi \text{ play } T^{-1}(C_n).$$

This set is Borel. It is also a legal response in the X -game. Indeed, let $L = a + \mathbb{R}x_n$ be an affine line in X parallel to x_n . Since $Tx_n \neq 0$, the restriction $T|_L$ is a nonconstant affine map from L onto the affine line $Ta + \mathbb{R}Tx_n$. Therefore

$$L \cap T^{-1}(C_n) = (T|_L)^{-1}(C_n \cap (Ta + \mathbb{R}Tx_n))$$

has one-dimensional Lebesgue measure zero on L , because $C_n \in \mathcal{A}_Y(Tx_n)$. Thus $T^{-1}(C_n) \in \mathcal{A}_X(x_n)$.

Now consider any legal play by Player I in X , so that (x_n) is dense in X . Since T is a continuous surjection, (Tx_n) is dense in Y . Removing zero terms leaves a dense sequence in Y : if the nonzero terms were not dense, some nonempty open subset of Y would contain only zero projected moves, impossible unless $Y = \{0\}$, where the conclusion is trivial.

The nonzero projected part of the X -play is therefore a legal play of the game in Y against the strategy Φ . Since Φ wins on B ,

$$B \subset \bigcup_{n:Tx_n \neq 0} C_n.$$

Taking inverse images gives

$$T^{-1}(B) \subset \bigcup_{n:Tx_n \neq 0} T^{-1}(C_n),$$

which is covered by Player II's moves in the X -game. Hence Ψ is a winning strategy for Player II on $T^{-1}(B)$.

For Corollary 1, apply the finite-dimensional positive case from the earlier packet: in a finite-dimensional real Banach space, every Borel Lebesgue-null set is II-winning in the Aronszajn-null game. The pullback theorem then gives the claim for $T^{-1}(N)$. Countable unions follow from Duda–Tsaban's lemma that II-winning sets form a Borel σ -ideal. $\square$

Limitations

This is not a full solution of the Duda–Tsaban conjecture. It covers finite-dimensional cylindrical null sets and their countable unions. A general Aronszajn-null set need not be known to decompose into such cylindrical pieces, and exactly this missing decomposition is close to the reverse-arrow problems in Duda–Tsaban's diagram.

Literature and Novelty Check

The source arXiv paper and the published article state the conjecture and the known chain

$$\sigma\text{-directionally porous} \implies \mathcal{C}^* \implies \text{II-winning} \implies \text{I has no winning strategy} \implies \mathcal{A}.$$

The previous lane-0 packet proved the finite-dimensional positive case. This packet adds the quotient-pullback permanence lemma and its cylindrical finite-dimensional corollary.

Searches performed for this push included the local run indexes; exact local source searches for “Aronszajn-null game”, “Duda Tsaban”, and “Conjecture 2.4”; web searches for the same phrases; Semantic Scholar citation lookup for arXiv:math/0601556; Crossref lookup for doi:10.1016/j.topol.2008.04.009; and Crossref/Springer metadata checks for Bogachev’s 2018 survey *Negligible Sets in Infinite-Dimensional Spaces*. These found no later paper claiming to solve the game conjecture. Crossref records one indexed citation of the published Duda–Tsaban article, while Semantic Scholar reports only Bogachev’s survey records.

References

- J. Duda and B. Tsaban, *Null sets and games in Banach spaces*, Topology Appl. **156** (2008), 56–60; arXiv:math/0601556.
- M. Csornyei, *Aronszajn null and Gaussian null sets coincide*, Israel J. Math. **111** (1999), 191–201.
- V. I. Bogachev, *Negligible Sets in Infinite-Dimensional Spaces*, Analysis Mathematica **44** (2018), 299–323.

Human Review Recommendation

Review should focus on the pullback legality argument and on the claim that nonzero projected directions remain dense after deleting zero projections. The corollary then follows formally from the finite-dimensional packet and the source paper’s sigma-ideal lemma.